

ZIHAN ZHANG

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EDUCATION

Smith College, Northampton, MA | Sept 2025 – Present

- **Candidate for Bachelor of Arts**
- **Major:** Neuroscience & Computer science **GPA: 3.96/4.0**
- **Relevant Coursework:** Fundamentals of Neuroscience, Clinical Neuroscience, Genomes & Genetics, Molecules, Cells & Systems, Linear Algebra, Introduction to Data Science, Introduction to Computer Science

Southwest Weiyu High School, Shanghai, China | Sept 2022 – June 2025

RESEARCH & ACADEMIC EXPERIENCE

Barresi Lab, Smith College | *Research Assistant* | Nov 2025 – Present

- Design and execute experiments investigating meteorin/meteroin-like morpholino influence on radial glial cell development in zebrafish spinal cord to characterize neural stem cell behavior.
- Apply techniques such as HCR, immunostaining, and confocal microscopy to quantify spatial gene expression patterns.
- Analyze fluorescent imaging to interpret developmental results (~8 hours/week)
- Represented the Meteorin team in the lab at Smith College's annual *Celebrating Collaborations* poster panel, presenting the team's CNS research findings to a cross-disciplinary audience of students, faculty, and prospective students.

Data Infrastructure Lab, Smith College | *Research Assistant* | Jan 2026 – Present

- Conduct Android reverse engineering to investigate app's logic using terminal-based development tools.
- Perform coding analysis to identify data privacy vulnerabilities and technical inconsistencies.
- Co-authoring an academic publication (in preparation) based on original research findings.

Summer Undergraduate Research Fellowship (SURF), Smith College | Research Fellow | May 2026 – Jun 2026

- Investigated muscle development in Metrn/MetrnL-deficient zebrafish using immunostaining and fluorescence microscopy.
- Performed whole-mount and cryosection immunostaining for myotendinous junction (MTJ) and neuromuscular junction (NMJ) markers.
- Collected and analyzed fluorescence images using Fiji and R to quantify muscle fiber orientation.
- Assisted with zebrafish embryo collection, fixation, tissue preparation, and imaging.

Shanghai Pudong Software Technologies Service Co., Ltd. | Project Intern, Innovation Division | Jun 2026

- Assisted with an AI-based river inspection project for environmental monitoring.
- Organized technical indicators for identifying visual abnormalities, including moss growth, pipeline leakage, and water stains.
- Participated in discussions with the Research and Development team and prepared project documentation.

Sungkyunkwan University Summer Program | Research Project | Jul 2026

- Investigated the relationship between EEG signals and consumer purchase intention through a hypothesis-driven research project.
- Collected and analyzed data using Python, following the complete data analysis pipeline, including data acquisition, preprocessing, statistical analysis, and visualization.
- Applied LLM-assisted text analysis and statistical methods to support data interpretation.
- Wrote a research paper and presented the project.

BIO 133 – Molecules, Cells, Systems, Smith College | *Teaching Assistant* | Jan 2026 – Present

- Prepare reagents, materials, and experimental setups for undergraduate biology labs
- Assist students during lab sessions with experimental techniques and biological concepts (6-7 hours/week)

Center for Systems Biomedicine, Shanghai Jiao Tong University | *Research Assistant* | Jan 2024 - Feb 2024

- Assisted with 293T-cell virus enveloping experiments, plasmid amplification, extraction, transformation, and transfection.
- Applied PCR and DNA gel electrophoresis procedures to verify genetic components for team research.

Biostatistics Analysis (Harvard University / Prof. Sam Richardson) | *Participant* | Jan – May 2023

- Applied linear regression and probability distribution theories to hospital longevity data using R Studio.
- Collaborated on a team project analyzing correlations between economic conditions and health factors in Uruguay.

Inspirit AI Scholars | *Participant* | Mar – June 2024

- Developed and compared performance of Sequential, CNN, and Transfer Learning models for automated pneumonia detection.
- Applied data augmentation techniques and analyzed potential racial and gender biases in AI models.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

International Student Organization (ISO) | *Social Media Coordinator* | Feb 2026 – Present

- Design posters, program leaflets, and interview videos for events such as Rhythm Nations
- Document events through photography and organize media files on Google Drive
- Manage Instagram campaigns to enhance campus engagement.

Student Union, Southwest Weiyu High School | *President of News Department* | May 2024 – May 2025

- Managed photography, video production, manuscript editing, and press coverage for school events.
- Co-founded the "The Diary" photo collection sale, coordinating across five international schools to raise funds for student charity programs.
- Organized the First Short Film Festival in school, producing promotional video, managing online submission platforms and judging coordination, and hosting the award ceremony.

Roots & Shoots “Organic Farming” Club | *President* | Sept 2023 – May 2025

- Organized monthly activities including campus biodiversity documentation, creating eco-friendly products, and vegetable cultivation.
 - Coordinated team members to complete projects on environmental awareness.
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SKILLS

- **Laboratory:** ZEISS ZEN Software, HCR, Immunostaining, PCR, Gel Electrophoresis, Plasmid Transfection
- **Technical:** Python (Data Analysis), R Studio, MatLab, Shell Scripting, Git.
- **Design & Media:** Adobe Photoshop, Illustrator, Premiere Pro, Canva, Digital Photography.
- **Languages:** Mandarin (Native), English (Fluent), Spanish & Japanese (Interests)